

HRHuman: Tuning-Free Higher-Resolution Human Image Generation via Template Knowledge

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Fig. 1: Higher-resolution human image generation is showcased at actual resolution of 2048^2 and 4096^2 . Built on SDXL, which supports generation up to 1024^2 , **HRHuman** seamlessly generates diverse human-centric images at $4\times$, $16\times$ scales, providing well-structured, highly detailed visuals. Please zoom in for a closer view.

Abstract—High-resolution human-centric image generation offers significant potential across various industries, such as entertainment, media, and fashion. Diffusion models for text-to-image generation have significantly improved the quality of human image synthesis. However, when scaling to higher resolutions (2K, 4K, and above), they often encounter issues such as object repetition and structural distortion, which appear especially unnatural in human images. To address these challenges, we propose HRHuman, a tuning-free framework for Higher-Resolution Human Image Generation. By leveraging an open-source large human vision model that incorporates rich template knowledge as prior, we first introduce the Prompt Discretization scheme to discretize user-input text prompts, mapping them to image elements and human body parts. Additionally, we implement a Template-guided Prompt Filtering mechanism to align these

discretized prompts with regional image semantics, ensuring fine-grained prompt guidance. Extensive experiments demonstrate that HRHuman achieves state-of-the-art performance in human-centric higher-resolution image generation, significantly addressing both issues of object repetition and structural distortion.

Index Terms—Human-centric Image Generation, Higher-resolution Generation, Tuning-free, Human Vision Model

I. INTRODUCTION

Advancements in generative models, particularly diffusion-based methods [4], [18], [19], [22], [26], [28], have transformed image generation by enabling highly realistic visuals from text prompts. Human-centric image generation, with applications in entertainment, virtual reality, healthcare, and

fashion, is a key beneficiary. While these models [3], [13], [14], [16], [25] excel at synthesizing detailed human figures with creative flexibility, generating high-quality images at 2K, 4K, and higher resolutions remains a challenge. Common artifacts such as object repetition, structural distortion, and loss of texture detail become more prominent, adversely affecting the overall image quality [6], [9], [10], [12], [21]. Moreover, developing tuning-free methods [1], [5], [10], [17], [24] for high-resolution human-centric image generation is essential to streamline the process without manual adjustments. Scalable solutions can lower barriers, making advanced image-generation technology more accessible.

To address these pressing challenges, we introduce HRHuman, a novel tuning-free framework for Higher-Resolution Human Generation. It leverages human template knowledge from large-scale human vision models as a prior, providing a solid foundation for human-centric image synthesis. HRHuman incorporates two key innovations: First, we introduce the Prompt Discretization scheme, which discretizes user-input text prompts and maps them to image elements and human body parts such as face, legs, and hands. Second, we implement a Template-guided Prompt Filtering mechanism to align these discretized prompts with corresponding regional image semantics, ensuring fine-grained prompt guidance in local patch-wise denoising and maintaining structural coherence during upscaling. Crucially, HRHuman operates as a tuning-free framework without any model parameter tuning, making high-quality, higher-resolution human image generation accessible. As illustrated in Figure 1, HRHuman produces high-quality images at various resolutions, effectively resolving issues of object repetition and maintaining detailed and coherent structures, even when zoomed in.

The contributions of this work are threefold: (1) We introduce HRHuman, a tuning-free framework for higher-resolution human image generation, utilizing human template knowledge as a prior and discretizing text prompts into distinct segments. (2) We propose a template-guided prompt filtering mechanism to align discretized prompts with regional image semantics, enabling fine-grained prompt guidance during patch-wise local denoising. (3) Extensive quantitative and qualitative evaluations are conducted to compare HRHuman with state-of-the-art methods, demonstrating the effectiveness of HRHuman.

II. RELATED WORK

Text-Driven Human-centric Image Generation. Recent advancements in this field focus on enhancing realism, coherence, and resolution of human figures. Make-a-Scene [7] proposes a hierarchical model that integrates background and foreground elements, improving human-object interactions and reducing pose misalignment artifacts. HumanSD [13], building upon the pre-trained Stable Diffusion (SD), focuses on highly precise pose and text-conditioned human image generation, targeting the accuracy of human depictions. More recently, CosmicMan [16] introduces a multi-scale diffusion approach tailored for human-centric image generation, substantially im-

proving the structural integrity and visual quality of human figures, though limited to around 1K resolution.

Tuning-free Higher-Resolution Image Generation. Recent developments in tuning-free higher-resolution image generation have focused on simplifying the generation process while achieving superior image quality. ScaleCrafter [10] introduces a novel approach to high-resolution image synthesis, emphasizing scale-aware feature crafting to achieve ultra-detailed images without manual tuning. Similarly, DemoFusion [5] proposes a diffusion-based framework that eliminates the need for fine-tuning. It leverages pre-trained models to generate high-resolution images while balancing computational efficiency and image fidelity. Multidiffusion [1] proposes a multistage diffusion process that progressively refines the generated image. FouriScale [12] offers a unique fusion strategy from a frequency perspective, concentrating on operations within convolutions to optimize image synthesis.

III. PRELIMINARIES

Latent Diffusion: The recent text-to-image diffusion model, specifically stable diffusion (SD) [19], [22], leverages a learned low-dimensional latent space for efficient diffusion and denoising processes. An autoencoder is employed for perceptual compression, which significantly reduces computational cost. The encoder E converts the image $I \in \mathbb{R}^{3 \times H \times W}$ into its latent representation $z_0 \in \mathbb{R}^{4 \times H' \times W'}$ as $z_0 = E(I)$, while the decoder D reconstructs the image as $\hat{I} = D(z_0)$.

Next, the diffusion model defines a fixed forward diffusion process that gradually adds noise to the latent code $z_0 \sim p(z_0)$:

$$q(z_t|z_0) = \mathcal{N}(z_t; \sqrt{\alpha_t}z_0, (1 - \alpha_t)\mathbf{I}). \quad (1)$$

During inference, latent features are sampled from the conditional distribution $p(z_0|P)$ using user-input text prompt P :

$$p_\theta(z_{0:T}|P) = p(z_T) \prod_{t=1}^T p_\theta(z_{t-1}|z_t, P). \quad (2)$$

MultiDiffusion [1]: built upon Latent Diffusion Models (LDMs), it introduces an overlapped patch-based denoising strategy to create high-resolution panoramas. Within the defined latent space $\mathbb{R}^{C \times H \times W}$, a sliding window mechanism samples a series of local latent representations at any arbitrary denoising step t , i.e., $Z_t^{\text{local}} = [z_{0,t}, \dots, z_{n,t}, \dots, z_{N,t}]$, where $z_{n,t} \in \mathbb{R}^{c \times h \times w}$ represents the latent space of the n -th local patch at step t . These local representations Z_t^{local} are then concatenated with overlapped regions averaged and eventually decoded into a high-resolution panoramic image.

IV. PROPOSED METHOD

A. Prompt Discretization for Human Template

As previously discussed, tuning-free higher-resolution image generation often encounters object repetition and structural distortion issues, largely due to insufficient or improper prompt guidance for local areas. While patch-wise parallel denoising [1], [5] provides an effective solution, mismatches between local features and textual prompts can still lead to undesired

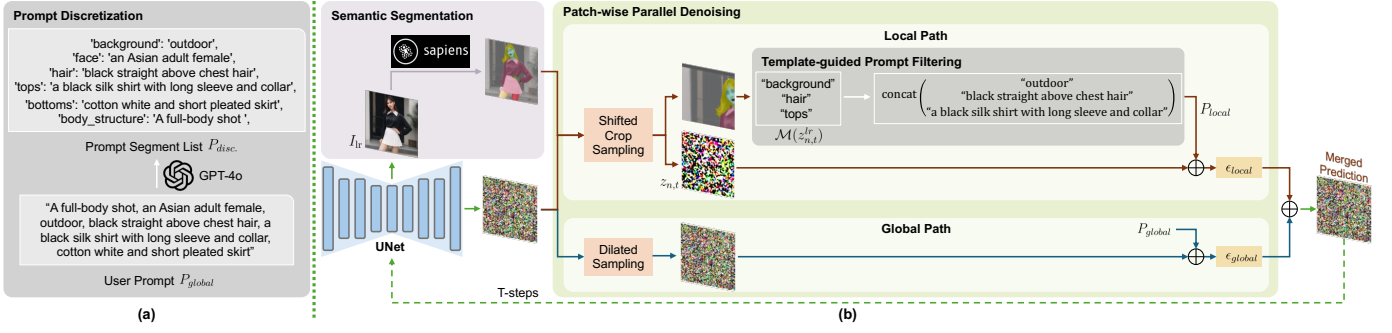


Fig. 2: The overall framework of **HRHuman**. (a) A user prompt P_{global} is decomposed using GPT-4o to a list of prompt segments P_{disc} accordingly in the Prompt Discretization scheme. (b) Sapiens first outputs a segmentation map using a low-resolution image I_{lr} as input. The patch-wise parallel denoising of HRHuman are depicted in the green box: local path and global path are employed in parallel to preserve local fine details and global coherent structure. Looking at the local patch denoising, both local latent representation and segmentation map are sampled by shifted cropping. The cropped semantic segmentation is further used to filter and sort the corresponding prompt segments $P_i \in P_{\text{disc}}$ through $\mathcal{M}(z_{n,t}^{lr})$. A new local prompt P_{local} is then sequentially concatenated to guide the local patch denoising stages. Dilated sampling to enable wider global context in global denoising stages. Local noise ϵ_{local} and global noise ϵ_{global} are merged for further denoising process.

outputs. Unlike general higher-resolution image generation, these issues can be potentially mitigated in human-centric generation scenario. Specifically, such prompts typically detail body regions, poses, clothing, and sometimes, backgrounds. In-depth descriptions are crucial for generating photorealistic and controllable images. These prompts are modular and can be truncated, allowing them to be selectively applied during patch-wise parallel denoising where appropriate.

To this end, the Prompt Discretization scheme is proposed to interpret the content of user-input text prompts and discretize them into distinct segments, each representing unique image elements, such as the background and specific human body parts. Specifically, we adopt a widely recognized human part-segmentation labeling scheme inspired by LIP (Look Into Person) [8], which utilizes $K = 20$ labels to categorize the background and 19 semantic human parts, including the head, upper arms, hair, and skirt. Intuitively, corresponding to K labels, a user-input prompt P_{global} can be easily discretized into a list of prompt segments P_{disc} :

$$P_{\text{disc}} = \{P_1, P_2, \dots, P_K\}, \quad P_i \cap P_j = \emptyset \quad \forall i \neq j, \quad (3)$$

where each P_i represents a distinct prompt segment corresponding to a specific body region or image element. Leveraging the capabilities of advanced language models, P_{disc} can be generated effortlessly using GPT-4o model with user-input prompts and provided labels, as shown in Figure 2 (a).

During the local patch denoising process, prompt segments $P_i \in P_{\text{disc}}$, aligned with local image semantics, are extracted and concatenated to form a new prompt, P_{local} , ensuring precise guidance. More details are discussed in next section.

B. Template-guided Prompt Filtering

Following DemoFusion [5], SDXL [19] is employed as the base model to create a low-resolution image I_{lr} from the global prompt P_{global} . Leveraging established cross-attention mapping

between text and image modalities ensures strong alignment between the composition of I_{lr} and the prompt list P_{disc} .

To enhance the Prompt Discretization process, a sophisticated human vision foundation model Sapiens [15] is integrated. As depicted in Figure 2 (b), Sapiens processes I_{lr} to produce a detailed 28-class body-part segmentation, which is a superset of K classes from LIP [8]. This segmentation provides a fine-grained understanding of the image's semantic structure, essential for the next steps of human-centric image synthesis. During patch-wise local denoising, the latent patch $z_{n,t}$ and the segmentation patch are cropped simultaneously. Semantic labels $\mathcal{M}(z_{n,t}^{lr})$ within in the patch are sorted and extracted, which are then used to retrieve the corresponding prompt segments $P_i \in P_{\text{disc}}$. To enable more controlled and precise generation, prompt segments from P_{disc} are sequentially concatenated, following the prompting strategy of Stable Diffusion models. The resulting local prompt P_{local} , employed during local patch denoising stages, is formulated as:

$$P_{\text{local}} = \text{concat}(\{P_i\}), \forall i \in \mathcal{M}(z_{n,t}^{lr}) \text{ and } P_i \in P_{\text{disc}}. \quad (4)$$

Then, P_{local} guides the local denoising stages, formalized as:

$$p_{\theta}(z_{n,0:T}|P_{\text{local}}) = p(z_{n,T}) \prod_{t=1}^T p_{\theta}(z_{n,t-1}|z_{n,t}, P_{\text{local}}), \quad (5)$$

To further refine the construction of P_{local} , we implement two key techniques: Lower-bound thresholding can be applied to exclude less relevant prompt segments from P_{disc} , preventing confusion within the diffusion model. Moreover, by assigning variable weights to prompt segments, we can direct the diffusion model's attention toward critical aspects of the image, resulting in a more coherent and accurate generation.

V. EXPERIMENTS

A. Experiment Setup

We validate the performance of HRHuman based on the SDXL [19]. To assess its effectiveness, we conduct compara-

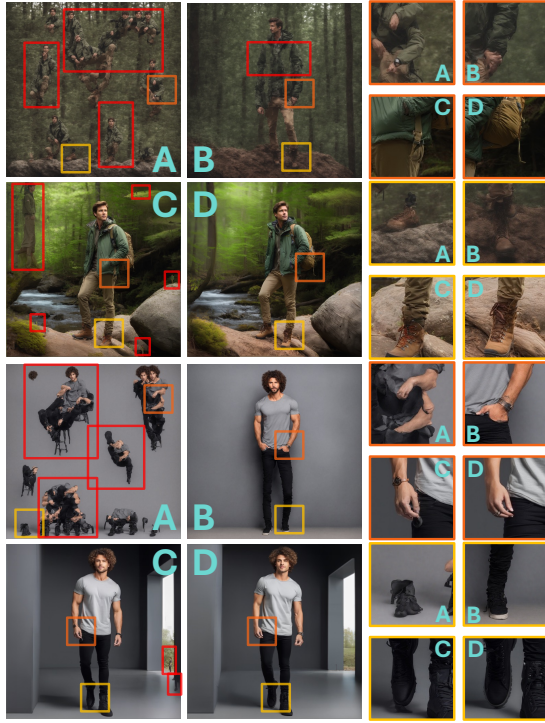


Fig. 3: Visual comparison across various image generation model: A: SDXL, B: ScaleCrafter, C: DemoFusion, and D: HRHuman, under the setting of 4096×4096 . The orange and yellow box highlight blurred and unreasonable elements, while the red box denotes the repeated object occurrence.

tive analyses against the following state-of-the-art tuning-free approaches: (i) SDXL Direct Inference (SDXL-DI) [19], (ii) SDXL+BSRGAN [27], (iii) ScaleCrafter [10], (iv) DemoFusion [5]. Following the same settings of prior methods [5], [10], we comprehensively evaluate the performance of our HRHuman model on $4\times$ and $16\times$ higher-resolution generation tasks, generating images at resolutions of 2048×2048 and 4096×4096 , respectively.

B. Qualitative Results

Figure 3 presents a visual comparison of 4096×4096 resolution images generated by HRHuman, SDXL-DI, ScaleCrafter, and DemoFusion. SDXL-DI tends to produce images with excessive object repetition, while ScaleCrafter struggles to maintain human body structure coherence. Enlarged views show texture details where the structure is reasonably intact. Despite DemoFusion incorporating global semantic information via residual connections and dilated sampling to preserve an accurate global structure, it continues to struggle with object repetition and local structural inaccuracies. In the first example, a man in a green jacket is generated where the trunk should be, with several smaller figures scattered over the leaves, stones, and moss. Enlarged views reveal visible artifacts on the bag strap and stacked trousers. In contrast, HRHuman generates a well-posed man in a complex forest background, with rich texture details on the jacket and

TABLE I: Comparison of quantitative metrics for different tuning-free high-resolution image generation methods at 2048^2 and 4096^2 resolutions. The best results are highlighted in **bold**, and the second-best are underlined.

Method	2048×2048 ($4\times$)			4096×4096 ($16\times$)		
	FID ↓	KID ↓	CLIP ↑	FID ↓	KID ↓	CLIP ↑
SDXL-DI	92.40	0.0333	16.31	224.76	0.1354	15.41
SDXL+BSRGAN	67.23	0.0186	<u>17.27</u>	67.16	0.0186	<u>17.35</u>
ScaleCrafter	67.79	0.0162	17.28	86.03	0.0288	17.56
DemoFusion	<u>62.14</u>	0.0130	17.19	<u>62.47</u>	<u>0.0124</u>	17.22
HRHuman (Ours)	62.04	<u>0.0133</u>	17.28	61.73	0.0123	17.29

backpack clearly visible in enlarged views. The left leg is positioned on a rocky surface, with the stacked trousers and boots exhibiting a well-defined structure and high fidelity. In the second example, DemoFusion’s result shows smaller figure repetition, inaccurate hand and shoe structures, and noticeable blurriness. Conversely, HRHuman generates a fit man with curly hair, dressed in a fitted gray t-shirt and black pants with well-formed hands and boot sneakers, and successfully eliminates the issue of mini-figure repetition.

C. Quantitative Results

For a fair comparison, we conduct quantitative experiments on LAION-5B [23], a large dataset comprising image-caption pairs. Following previous work [5], we randomly sample 1000 prompts from LAION-5B to create a human-centric subset that encompasses diversity in ethnicity and age, highlighting both global body structures and prominent facial features. For a robust quantitative comparison, we adopt three widely used evaluation metrics: Fréchet Inception Distance (FID) [11], Kernel Inception Distance (KID) [2], and CLIP Score [20]. As shown in Table I, HRHuman achieves significant performance gain over existing methods across most metrics. Notably, HRHuman outperforms DemoFusion in FID and KID for $16\times$ higher-resolution generation, while also achieving competitive CLIP evaluation scores. These demonstrate HRHuman’s superior capability in generating higher-resolution, structurally coherent human images, effectively addressing common issues like repetition and poor composition.

VI. CONCLUSION

In conclusion, HRHuman advances higher-resolution human-centric image generation by resolving issues like object repetition and structural distortion common in such tasks. Integrated with an open-source large human vision model, HRHuman introduces the Prompt Discretization scheme to map text prompts to specific image elements or human body parts, and the Template-guided Prompt Filtering mechanism for precise alignment in text segments and image semantics. Extensive experiments show HRHuman achieves state-of-the-art performance in human-centric image synthesis, providing a robust, tuning-free solution for generating high-fidelity images at resolutions of 2K, 4K, and beyond.

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